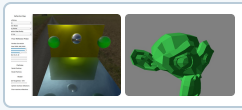
Study 1: 32 programs, checker frozen before its first run



Qt Quick 3D

14 programs · Qt 6.4.2

3 true causes ↑2 ●1

frozen		**8%**	88 / 12 / 1,026
observed		**88%**	88 / 12 / 0
release		**8%**	88 / 10 / 1,026 + 2 unres.

false alarms: host writes into per-frame slots smaller than a page (conservative capture)



wgpu

15 examples

0 true causes

frozen		**0%**	0 / 0 / 945
observed		**0%**	0 / 0 / 906
release	no report	–	0 groups

false alarms: three checker defects, fixed after the study



Dear ImGui

2 programs · 9d9a621f

1 true cause ●1

frozen		**100%**	40 / 0 / 0
observed		**100%**	40 / 0 / 0
release		**100%**	40 / 0 / 0



vkcube

1 program · 1.3.275

1 true cause ↑1

frozen		**100%**	2 / 0 / 0
observed		**100%**	2 / 0 / 0
release		**100%**	2 / 0 / 0

Study 2: 25 programs, a later checker frozen before its first run



GTK 4

8 + 1 programs · 4.14.5

3 true causes ↑3

frozen		**14%**	173 / 0 / 1,048 + 13 unres.
observed		**14%**	173 / 0 / 1,048 + 13 unres.
release		**92%**	173 / 1 / 6 + 8 unres.

false alarms: a frame's images in one descriptor array (frozen); atlas rectangles (release)



vkmark

7 scenes · ab6e6f3

5 true causes ↑2 ●3

frozen		**100%**	38 / 0 / 0
observed		**100%**	38 / 0 / 0
release		**100%**	38 / 0 / 0



mpv / libplacebo

4 configurations · 0.37 / 6.338

1 true cause ↑1

frozen		**100%**	16 / 0 / 0
observed		**100%**	16 / 0 / 0
release		**100%**	16 / 0 / 0



GStreamer

5 pipelines · 1.24.2

2 true causes ●2

frozen		**100%**	33 / 0 / 0
observed		**100%**	33 / 0 / 0
release		**100%**	33 / 0 / 0

true technicality unresolved false alarm bars: share of report groups; numbers: % true, then true / technicality / false alarm

frozen: the fresh result (conservative capture, the theorem's configuration) **observed**: same checker, observed capture *release* (dashed): post-hoc, same traces

↑ changed upstream before the study ● still present upstream